



M.SC. DATA ANALYTICS & TECHNOLOGIES

DAT7301 - DATA ANALYSIS AND VISUALISATION

ASSESSMENT 3 - A COMPREHENSIVE REPORT

STUDENT NAME: VICTOR ENE

STUDENT NUMBER: 2318217

MODULE TUTOR: DR EUGEN HARINDA

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## **Abstract**

This is a comprehensive report on the statistical analysis of UK seafood production and trade patterns. The data contained records from 2009 to 2020 and was aggregated from information obtained from the Marine Management Organisation (MMO), HM Revenue and Customs (HMRC) and the Centre for Environment, Fisheries and Aquaculture Science (CEFAS). After preparing and exploring our data, Key findings from correlation testing revealed no strong correlation between domestic production and trade volumes of seafoods with a persistent seafood trade deficit. Forecasts into the production of seafoods using ARIMA and Holt's Linear ETS models suggested a stable increase in production. The insights drawn from the report can be valuable in supporting decisions in supply chain management of seafoods, fiscal policies and supporting investors in their planning.

## **Keywords**

Seafood, Production, Imports, Exports, Trade, ARIMA, ETS, Forecasting, Data Analysis, Visualisation

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## **1. Executive Summary**

This report examines the UK seafood industry by analysing the relationship between domestic production and international trade, identifying dominant species in UK seafood trade and forecasting future production volumes of seafood. The analysis is based on data from 2009 to 2020, sourced from reputable organisations. The analysis revealed a consistent trade deficit, with imports exceeding exports throughout the period. Correlation analysis indicates minimal association between production and trade volumes. Time-series forecasts suggest a stable to slightly increasing production trend. These findings offer valuable insights for policymakers and industry stakeholders in making informed decisions regarding managing resources, trade strategies and future planning.

## **2. Methodology**

### **2.1 Data Sources**

The dataset for this analysis covers the period from 2009 to 2020 and was compiled from multiple credible sources: Marine Management Organisation, HM Revenue and Customs and Centre for Environment, Fisheries and Aquaculture Science.

### **2.2 Study Design and Data Pre-processing**

The study adopted a quantitative approach, applying statistical analysis and time-series forecasting to explore the relation between UK seafood production and trade and forecast production. Data preparation was carried out in R, where libraries such as tidyverse, forecast, and ggplot2 were first loaded. The dataset "UKSeafoodSupplyChain.csv" was then imported and missing production volumes were imputed using species-specific means to maintain analytical consistency. Irrelevant columns were removed to work with relevant data. After converting volumes from grams to tonnes for easier interpretation, the dataset records were filtered to retain only production, imports, and exports, excluding purchases and consumption. Finally, the data was checked for duplicates to ensure integrity before commencing analysis.

### **2.3 Analysis Techniques**

Several analytical techniques were applied during the study. Exploratory Data Analysis (EDA) was used to generate summary statistics and visualise data distributions. Pearson correlation analysis helped assess the strength of relationships between UK seafood production and trade volumes. For forecasting future production volumes, time-series models such as ARIMA and Holts Linear Exponential Smoothing (ETS) were used to project production trends from 2021 to 2025

### 3. Key Findings and Insights

#### 3.1 Exploratory Data Analysis

Exploratory data analysis of UK seafood data from 2009 to 2020 revealed some clear trends. Production rose from around 6,200 tonnes in 2009 to a peak of 7,550 tonnes in 2014, before falling back to roughly 6,500 tonnes by 2020. Imports and exports fluctuated more noticeably. Imports began at approximately 13,700 tonnes in 2009 and declined steadily to about 11,000 tonnes by 2015. Imports experienced a modest rise in 2016 to around 11,500 tonnes, before stabilizing near 11,200 tonnes through to 2020. Exports were more volatile, starting at around 8,700 tonnes in 2009, increasing slightly in 2010, then dropping to 7,100 tonnes by 2013. Recovering to about 8,050 tonnes in 2014, but declined again to around 6,700 tonnes by the end of the period.

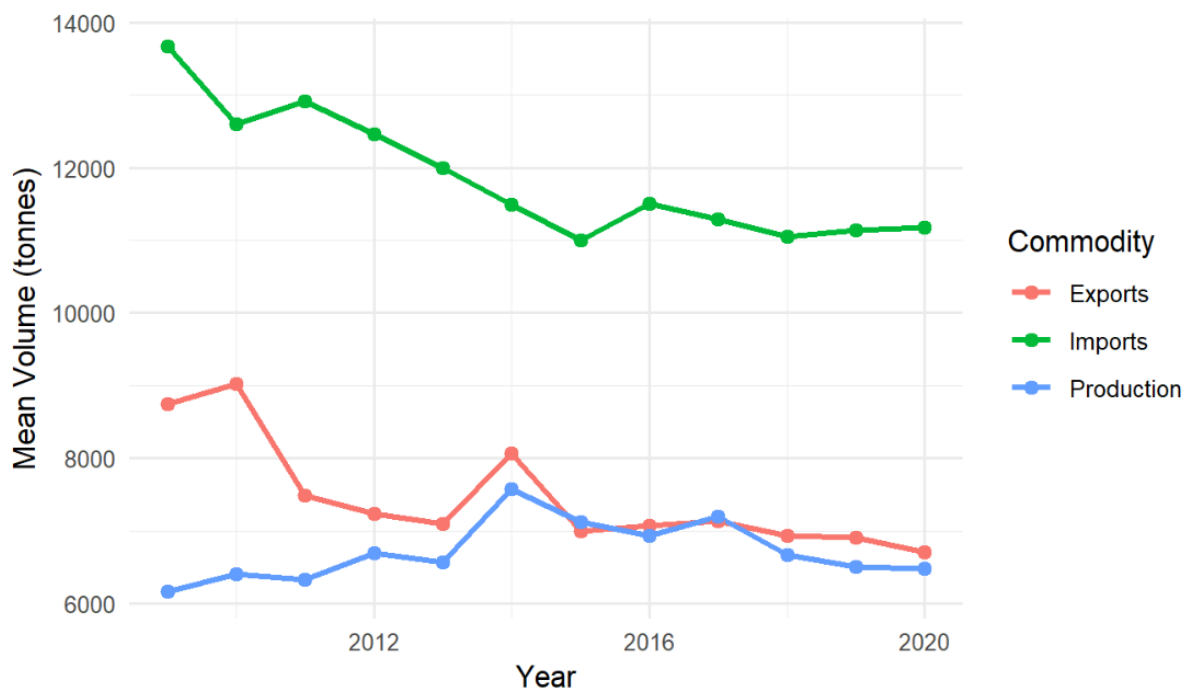
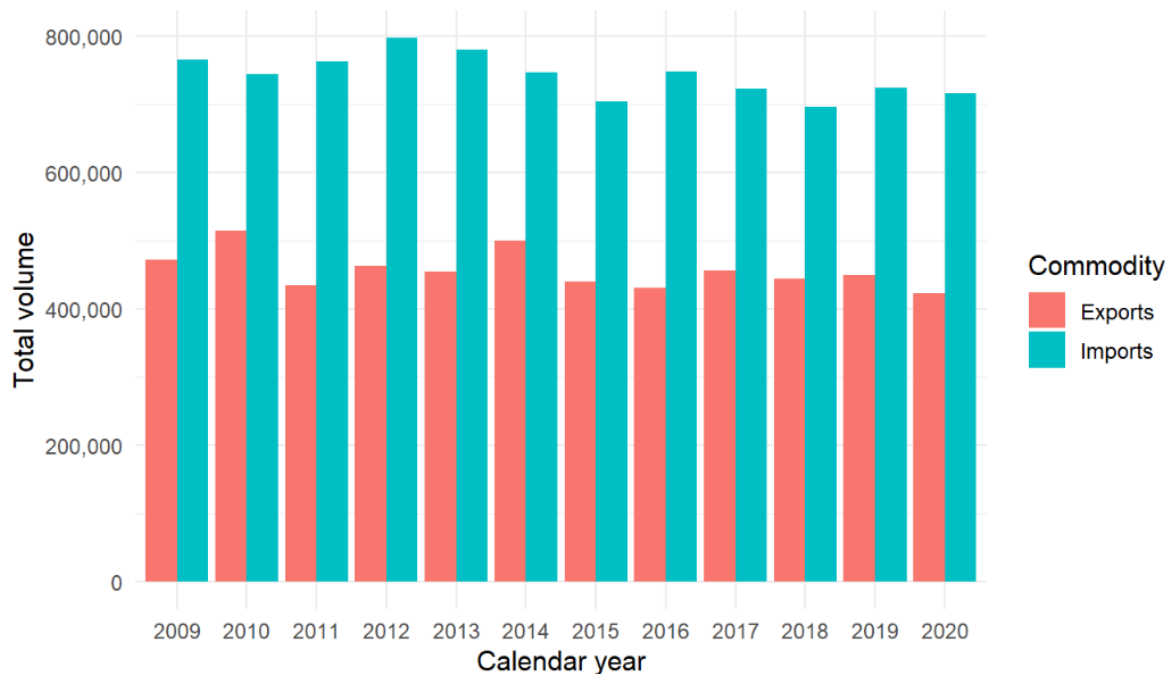


Figure 1 Seafood trade volume

### 3.2 UK Seafood Trade Balance

The UK seafood trade imports consistently surpassed exports every year throughout the period (2009 to 2020). This pattern reflects the reliance on imported seafood to meet domestic demand. The imports volume reached its highest point in 2012, with approximately 800,000 tonnes. In contrast, the peak for exports occurred earlier in 2010, when around 520,000 tonnes were shipped abroad. The most trade imbalance was observed in 2012, when imports exceeded exports by roughly 330,000 tonnes. This significant gap highlights the UK dependency on external seafood sources and also raises important questions around food security and opportunities for strengthening domestic production. Such a consistent trade deficit suggests that future policy and investment decisions may need to focus on enhancing local supply chains and reducing over reliance on imports where feasible.



*Figure 2 import and Export*



3.3 Dominant Import and Export Species

Prawns account for the largest share of UK seafood imports, with total volumes exceeding 1.6 million tonnes over the period showing the UK strong dependence on imported shellfish. Tuna and cod also represent major import commodities, driven by high consumer demand alongside limited domestic supply. On the export side, salmon stands out as the UK leading seafood export, followed by mackerel and herring, reflecting the strengths in cold water fish production for international markets. Salmon is also largely imported despite the UK being well established in the salmon farming industry, suggesting that imported varieties still play a role in meeting the market preferences or seasonal demand.

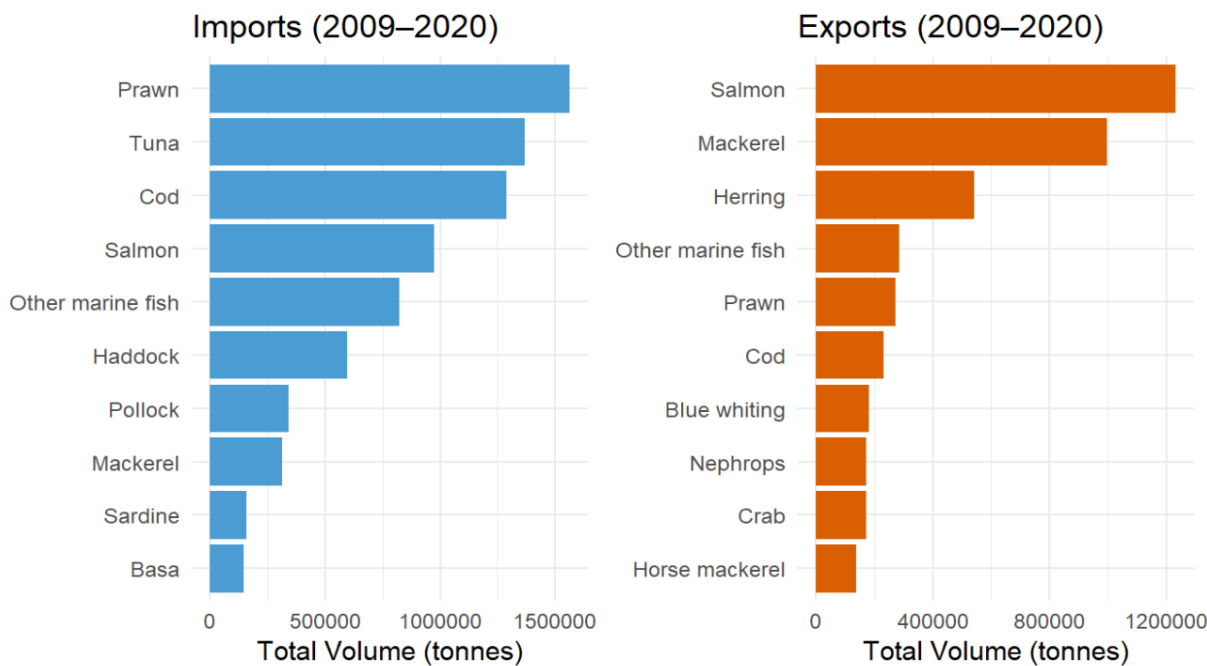
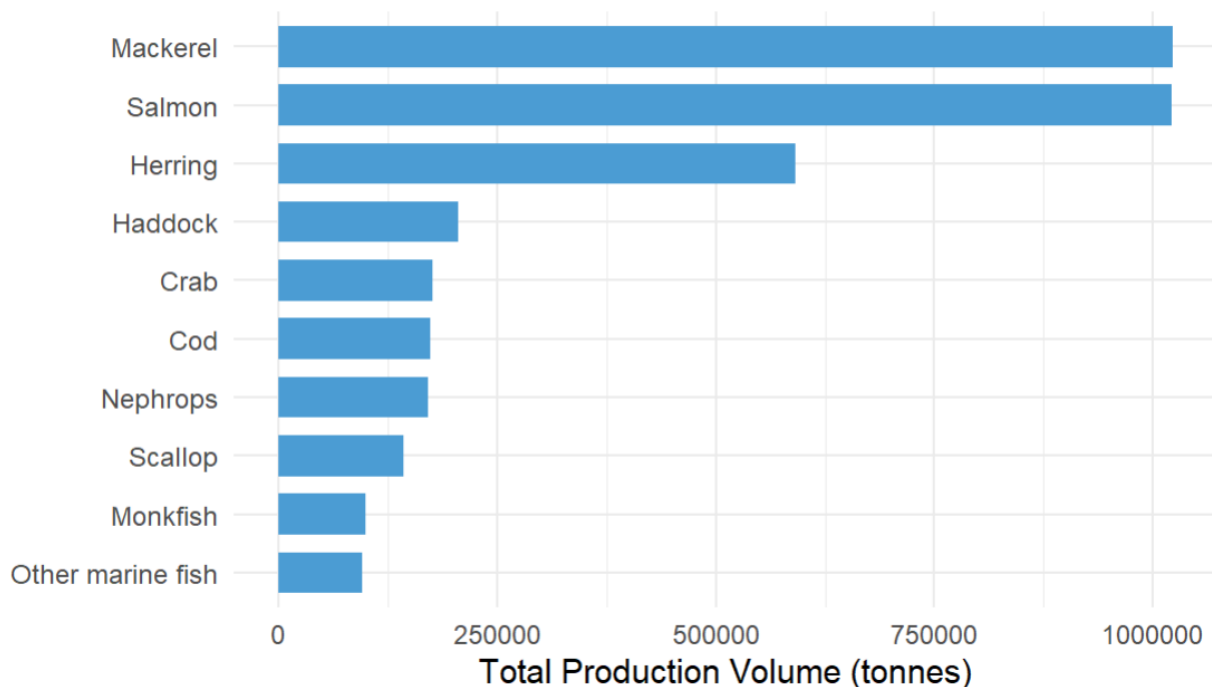


Figure 3 Seafood Species

### 3.4 UK Seafood Production

Mackerel and salmon dominate UK seafood production, with each species exceeding one million tonnes over the period. Herring ranks third in overall production volume. Other notable species include haddock, crab, cod, and nephrops, representing a diverse mix of demersal fish and crustaceans.



*Figure 4 Total trade*

### 3.5 Correlation Between Seafood Production and Trade

The correlation analysis conducted indicated that UK seafood production operates largely independently of international trade. Specifically, with a Pearson correlation coefficient of -0.08 and a p-value of 0.81 there was no statistically significant linear relationship between production and the combined volume of imports and exports. Similarly, when assessed individually, imports and exports also showed weak and statistically insignificant correlations with production, yielding

coefficients of -0.09 and 0.15 respectively. These findings imply that the UK's seafood trade dynamics may be shaped more by external market forces.

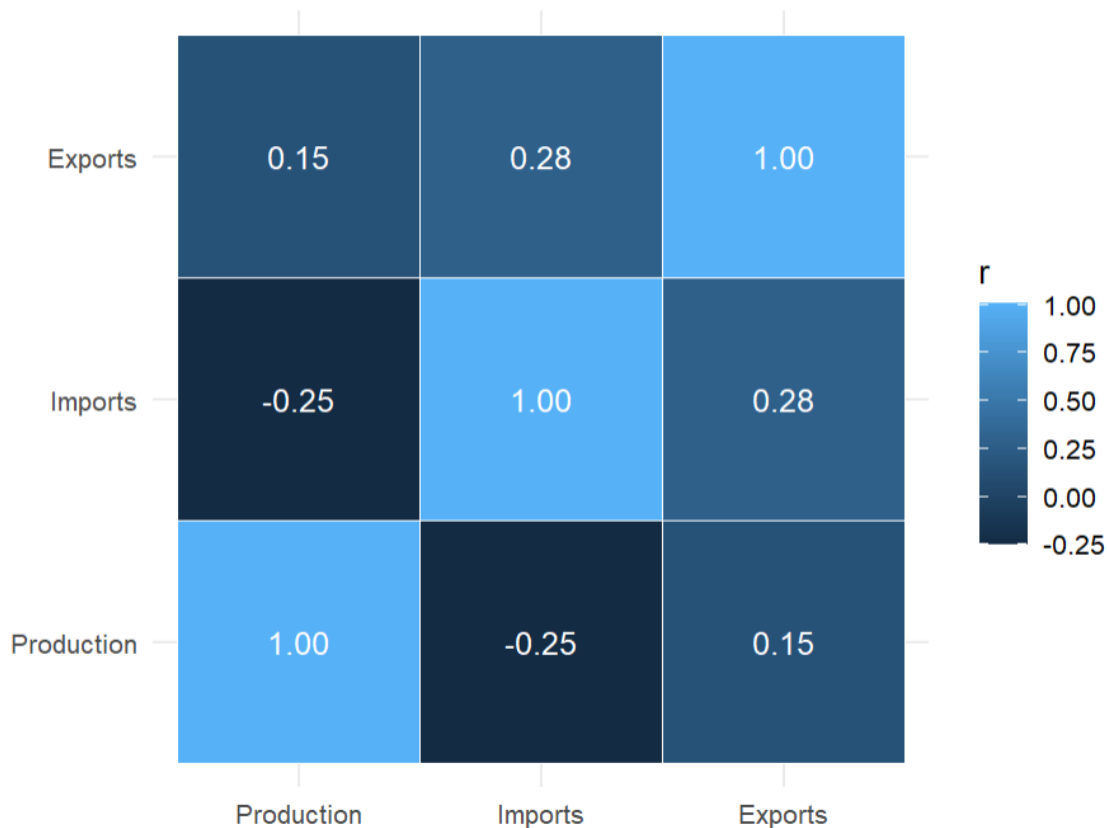


Figure 5 correlation matrix

3.6 UK Seafood Production Forecast

Time-series forecasting using the ARIMA model suggests that UK seafood production will remain relatively stable between 2021 and 2025, with projected volumes holding steady at around 390,000 tonnes per year. In contrast, Holt’s Linear Exponential Smoothing (ETS) model indicates a modest upward trend over the same period, with production expected to increase gradually from approximately 385,000 tonnes in 2021 to about 405,000 tonnes by 2025. These different projections highlight the sensitivity of future estimates to model assumptions, and suggest that

while sharp changes are unlikely, a slow and steady growth is possible depending on broader sector conditions.

| Year | ARIMA Forecast (tonnes) | Holt's Linear ETS Forecast (tonnes) |
|------|-------------------------|-------------------------------------|
| 2021 | 390,000                 | 385,000                             |
| 2022 | 390,000                 | 390,000                             |
| 2023 | 390,000                 | 395,000                             |
| 2024 | 390,000                 | 400,000                             |
| 2025 | 390,000                 | 405,000                             |

*Figure 6 Model values*

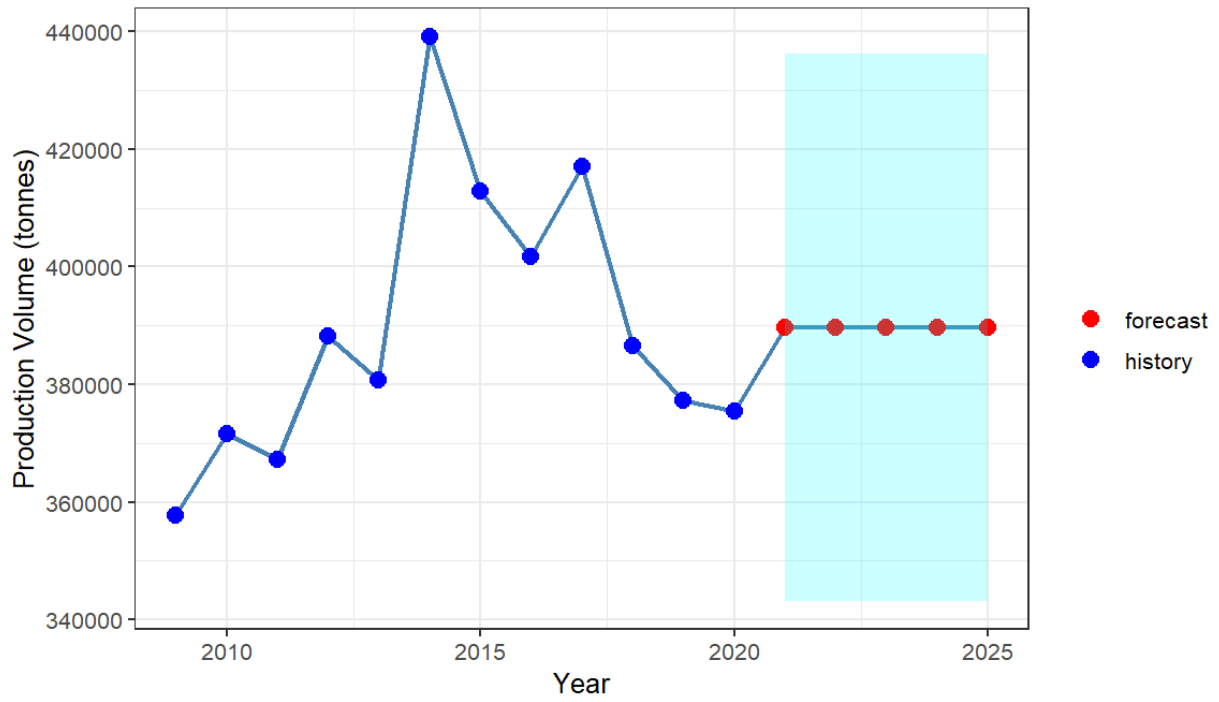


Figure 7 Arima forecast

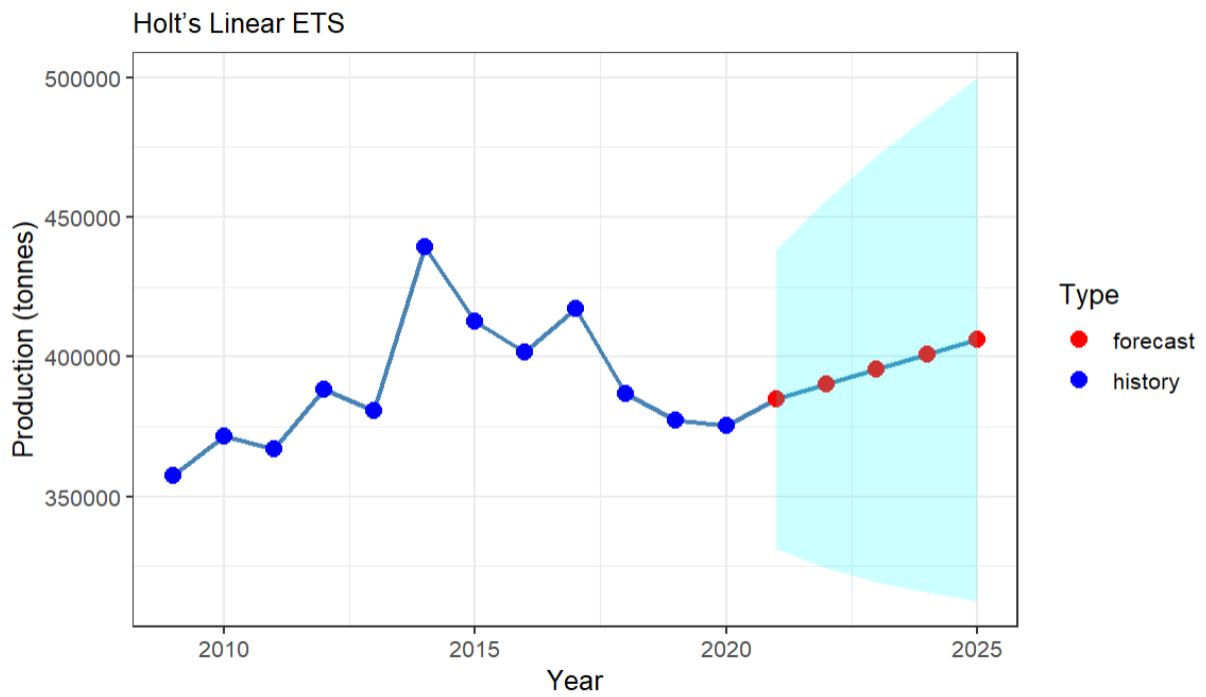


Figure 8 Holts Linear forecast

#### 4. Ethical Considerations

The analysis of seafood production and trade data involves several ethical responsibilities. First and foremost is the accuracy and reliability of the data used, the data was from trusted sources such as the Marine Management Organisation (MMO), HM Revenue & Customs (HMRC) and the Centre for Environment, Fisheries and Aquaculture Science (CEFAS) which helped ensure data integrity. Nevertheless, it is essential to recognise potential limitations, including possible inaccuracy or biases from the original data collection processes and also avoiding exaggeration and acknowledging assumptions made in the analysis and uncertainties within forecasting models and statistical inferences.

Furthermore, considering of how these findings may affect stakeholders. For instance, identifying dominant species in trade flows could influence policymaking around imports, exports and resource allocation. As such, results should be communicated responsibly and with sensitivity to the potential implications for different groups, including fishing communities, seafood processors and consumers.

## 5. Conclusion and Recommendations

This report examined UK seafood production and trade from 2009 to 2020. The analysis revealed a consistent trade deficit, with imports particularly prawns, tuna, and cod outweighing exports. Salmon, mackerel, and herring were the top export species. No significant correlation was found between domestic production and trade volumes and forecasting results suggest stable or slightly rising production through 2025.

The findings indicate a clear reliance on seafood imports, particularly for species such as prawns, which points to the need for properly managing international supply chains. The stable and slight decline observed in domestic production forecasts may assist with long term planning in the fishing sector. Additionally, since there was no significant correlation between production and trade volumes, it suggests that external factors play a more dominant role in shaping the UK seafood trade. Policymakers should consider strategies to increase exports by increasing production, which may also involve supporting domestic production and industry stakeholders.

### GAI Declaration

I confirm that no generative AI tool was used in the development of this assessment.